

Gustavo Valdivia-Mera

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ORCID | INSPIRE-HEP

Education

University of Houston Ph.D. in Physics	2023–Present <i>GPA: 4.0/4.0</i>
• Basic Core: Methods of Mathematical Physics, Advanced Mechanics, Quantum Mechanics I–II, Electrodynamics, Statistical Physics. • Advanced Core/Electives: Quantum Field Theory I–II, Advanced Statistical Mechanics, General Relativity and Cosmology, Quantum Information.	
ICTP-EAIFR / University of Rwanda M.Sc. in High Energy Physics	2020–2022 <i>Ranked 1st in cohort (HEP section)</i>
• Thesis: <i>Bounds on eV-scale sterile neutrinos from neutrinoless double-beta decay.</i> • Completed the ICTP Diploma Programme in High Energy Physics during the M.Sc. program.	
Universidad Nacional Mayor de San Marcos Professional Degree in Physics	2018–2019 <i>Distinction: Outstanding</i>
• Thesis: <i>On the Underlying Relation Between Quantum Entanglement and Wormholes.</i>	
Universidad Nacional Mayor de San Marcos B.Sc. in Physics	2012–2017 <i>Ranked 1st in cohort</i>

Research Experience

Hosur Group, University of Houston Research Assistant	2026–Present <i>Houston, TX</i>
• Study theoretical problems in strongly correlated quantum many-body physics, topological quantum matter, and anomalous transport.	
Ordóñez Lab, University of Houston Research Assistant	2023–2025 <i>Houston, TX</i>
• Investigated the emergence of thermality and information-theoretic structures in causally restricted spacetime domains, including causal diamond horizons and black holes.	
Theoretical Physics Group, UNMSM Undergraduate Research Affiliate	2015–2019 <i>Lima, Peru</i>
• Worked on the AdS/CFT correspondence, black hole thermodynamics, and the relation between entanglement and spacetime geometry.	

Publications

- **The Helical SYK Model and Emergent Infrared Integrability**
Gustavo Valdivia-Mera, Bhavay Tyagi, Eric R. Bittner, and Pavan Hosur
arXiv preprint. arXiv:2606.17161 (2026). [arXiv](#)
- **Thermal nature of the causal diamond horizon: A hidden property of the inertial propagator**
Nada Eissa, Carlos R. Ordóñez, and Gustavo Valdivia-Mera
Physical Review D. 112, 125008 (2025). [Journal](#) | [arXiv](#)
- **Horizon brightened acceleration radiation entropy in causal diamond geometry: A near-horizon perspective**
Nada Eissa, Carlos R. Ordóñez, and Gustavo Valdivia-Mera
Physical Review D. 112, 085030 (2025). [Journal](#) | [arXiv](#)
- **Path integral derivation of the thermofield double state in causal diamonds**
Abhijit Chakraborty, Carlos R. Ordóñez, and Gustavo Valdivia-Mera
Classical and Quantum Gravity. 42, 025015 (2025). [Journal](#) | [arXiv](#)
- **On the Unruh effect and the thermofield double state**
Gustavo Valdivia-Mera
International Journal of Modern Physics D. 34, 2530002 (2025). [Journal](#) | [arXiv](#)

Selected Presentations

Strange Metals and Black Holes: A Holographic Correspondence Research Day, Department of Physics, University of Houston, Houston, TX	<i>March 2026</i>
Thermal Nature of the Causal Diamond Horizon TINKUY 2026: “Un Panorama de la Física Teórica en el Perú”, Lima, Peru	<i>February 2026</i>
Towards HBAR in the causal diamond near horizon IQSE Summer School on Quantum Science, Texas A&M University, Casper, WY	<i>July 2025</i>
Near-horizon (conformal) aspects of black holes and the universality of HBAR entropy AMO/IQSE Seminar, Texas A&M University, College Station, TX	<i>April 2025</i>
Path integral derivation of the thermofield double state in causal diamonds APS Global Physics Summit, Anaheim, CA	<i>March 2025</i>
Quantum Algorithms I & II Quantum Scholars Program, QuantumQuipu, online	<i>August 2023</i>
Bounds on eV-scale sterile neutrinos from neutrinoless double-beta decay Summer School on Particle Physics, ICTP, Trieste, Italy	<i>June 2023</i>

Teaching Experience

University of Houston Teaching Assistant	2023–2025 <i>Houston, TX</i>
• Quantum Mechanics, undergraduate course	Fall 2025
• Quantum Mechanics II, graduate course	Spring 2025
• Physics Laboratory, undergraduate course	Fall 2023
TECSUP Instructor, General Physics	2018 <i>Lima, Peru</i>

Awards and Honors

• Student Ambassador, American Physical Society (APS)	2026–2027
• Lydia Mendoza Fellowship, University of Houston	2024–2026
• Presidential Fellowship, University of Houston	2023–2025
• 1st Place, Quantum Computing Fall Fest, University of Houston	2023
• Research Scholarship, UNMSM Vice-Chancellor’s Office	2018

Technical Skills

Python; Fortran; Mathematica; L^AT_EX.